Dr Andi Lowe

Data Scientist · PhD in Particle Physics · Ex-CERN · Breakthrough Prize Laureate

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PROFESSIONAL PROFILE

British data scientist with eight years of experience consulting for a diverse range of business clients and a 15-year background in data-driven scientific research within large international teams. Holds a PhD in particle physics, spent several years at CERN in Geneva, Switzerland, and was part of the team that discovered the Higgs boson. Co-recipient of the 2025 Breakthrough Prize in Fundamental Physics, awarded for ground-breaking measurements of Higgs boson properties and exploration of nature at the shortest distances and most extreme conditions. Expertise includes statistical data analysis, machine learning (ML), software development, mathematical modelling, data visualisation, and results interpretation. Co-author of over 500 peer-reviewed scientific publications and has presented at numerous international workshops and conferences.

EMPLOYMENT

2017–present **Data Scientist** *EPAM Systems Inc., Hungary*

Collaborated directly with business clients on diverse consulting projects, including:

- *Verizon Connect*
  Developed machine learning models for predictive maintenance of GPS fleet tracking devices using AWS Sagemaker, Python, SQL, and AWS Redshift to identify faulty units.
- *Texas Instruments*
  Developed a proof of concept for remote sensing using millimetre-wave radar technology. Conducted analysis and image processing with OpenCV to support project decision-making.
- *Child Mind Institute*
  Designed a ML algorithm in Python with Scikit-Learn to detect and monitor harmful body-focused repetitive behaviours, such as hair-pulling, skin-picking, and nail biting. Contributed to the development of a wrist-worn monitoring device to help healthcare professionals track treatment outcomes. Oversaw data collection methods and hardware selection, while identifying and mitigating project risks through effective team communication.
- *Merck Sharp & Dohme*
  Analysed neuromarketing data in Python to identify key features for an email marketing campaign.
- *London Stock Exchange Group*
  Developed and maintained software in R and Python to anonymise Personal Identifiable Information (PII) data, ensuring compliance with EU GDPR data protection laws. This safeguarded a critical $250 million annual revenue stream. The cloud-hosted, dockerised software uses ML to generate synthetic datasets with robust privacy protections for non-production environments. Established best practices for GDPR compliance and provided science-backed guidance on reducing the client's digital carbon footprint.
- *Bayer AG*
  Built extract, transform, and load (ETL) processes using KNIME to automate Excel financial report generation, reducing processing time from over 24 hours to under 2 hours.

2013–2017 **Research Fellow** *Wigner Research Centre for Physics, Hungarian Academy of Sciences, Hungary*

Conducted data analysis in R and C++ for the ALICE experiment at CERN, recreating conditions believed to have existed shortly after the Big Bang. Developed ML classification algorithms to identify particles based on their decay properties. Engaged with the data science community through public outreach talks and conference presentations.

2010–2012 **Postdoctoral Fellow, Deputy Team Leader** *California State University, Fresno, USA (based at CERN)*

Investigated the utility of hundreds of predictor variables for data-mining analyses using Monte Carlo simulations in C++. Engineered features to improve search sensitivity for new particles, reducing the time and cost of experimental data collection.

2008–2009 Postdoctoral Fellow *Indiana University, USA (based at CERN)*

Developed an algorithm in C++ and Python for real-time particle identification in streaming data at 1 GB/s. Optimised parameters to achieve high performance. This algorithm has been in production since 2010, processing tens of petabytes of data and supporting the ATLAS experiment's physics programme.

EDUCATION

- **PhD Particle Physics** *Royal Holloway, University of London, UK (including 1.5 years at CERN)*

  Contributed to the development of core software and algorithms in C++ for a real-time multi-stage cascade classifier, reducing collision event data rates from 60 TB/s to 300 MB/s for permanent storage. Conducted detailed time profiling and implemented optimisations that improved software performance by a factor of eight, meeting critical system requirements. Analysed petabyte-scale datasets using distributed computing systems. Developed software used in the discovery of the Higgs boson.

- **MSc Particle Physics** *Royal Holloway, University of London, UK*

  Investigated the search potential for the Higgs boson via the $H \to b\bar{b}$ decay channel using the ATLAS detector at CERN. Conducted the first data-mining analysis of this type entirely in C++, setting a benchmark for subsequent research in the field.

- **BSc (Hons) Physics** *Royal Holloway, University of London, UK*

TRANSFERABLE SKILLS

- **Communication:**
  Skilled at collaborating with clients as a consultant and effectively communicating with business stakeholders and technical teams to convey complex data-driven insights and technical specifications. Proficient in translating client requirements into detailed project documentation. Invited speaker at numerous international conferences. Experienced in visual storytelling and data visualisation best practices. Adept at simplifying complex technical concepts for diverse audiences. Recognised for building strong client relationships.

- **Problem solving:**
  Proven ability to lead independent research, analyse complex problems, and develop innovative, effective solutions.

- **Project management:**
  Experienced in managing multiple projects under strict deadlines, both onsite and remotely. Familiar with Agile and Waterfall software development methodologies. Skilled in leading project meetings and mentoring team members.

PROFESSIONAL CERTIFICATES

- **AWS Certified AI Practitioner**, Amazon Web Services (AWS), issued Nov. 2024.
  Foundations of artificial intelligence (AI), ML, generative AI, large language models (LLMs), prompt engineering, and AWS managed AI/ML services (*e.g.*, Bedrock, Amazon Q, SageMaker, *etc.*).

- **Salesforce Certified AI Associate**, Salesforce, issued Jan. 2025.
  AI foundations, AI capabilities in customer relationship management (CRM), ethical considerations of AI.

COMPUTING SKILLS

- **Programming languages:** Python (NumPy, Pandas, Scikit-Learn, Matplotlib, *etc.*), R, C++, Bash, SQL, Octave
- **Data mining software:** AWS Sagemaker, KNIME, Weka, XGBoost, H_2O, caret, ROOT
- **Data visualisation:** ggplot2, Plotly, TIBCO Spotfire, Tableau, Flexdashboard
- **Notebooks/Documentation:** Jupyter, Google Colab, R Markdown, LaTeX, Confluence
- **Software development:** Docker, Git/GitHub/GitLab, Jira, RStudio, UML, Valgrind, Visual Studio Code
- **Operating systems:** Unix, Linux, Microsoft Windows, macOS
- **Other:** object-oriented analysis and design, CPU and time profiling, code optimisation, memory debugging.